THANH VU

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RESEARCH INTERESTS

Machine learning, probabilistic reasoning, and applied statistics, especially honest model evaluation when data is noisy, biased, or shifting, and the data systems that make that evaluation reproducible.

EDUCATION

Franklin & Marshall College — Lancaster, PA

Bachelor of Arts in Computer Science | *GPA: 3.72/4.0 (Major: 3.82/4.0)* — May 2026

- Magna Cum Laude; Honors List (5 semesters); Dean's List (Spring 2022)
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Algorithms, Data Structures (C++), Theory of Computer Science, Information Theory, Calculus III, Linear Algebra, Discrete Math, Probability & Statistics, Applied Regression

RESEARCH & APPLIED PROJECTS

Can Mail Data Predict Housing Vacancy? | *Microsoft Fabric, Power BI* — Sep 2026

- Investigated whether quarterly USPS vacancy counts can stand in for the annual Census ACS vacancy rate, building a reproducible pipeline over 6 counties and 15 years (**84 usable county-year observations**).
- Evaluated it three ways (correlation, directional accuracy, bias): a pooled **r = 0.77** fell to between −0.01 and 0.72 within counties, and directional agreement was only **57.7%**, concluding USPS data is a county-dependent early signal, not a substitute.
- Flagged a 2012 USPS classification change that would otherwise have read as a false trend, and fixed two silent ingestion bugs.

Grocery Transaction Data Warehouse | *SQL Server, T-SQL* — Jul 2025 – Aug 2025

- Built a Medallion-architecture warehouse over **131M+ transaction lines** with stored-procedure ETL, a star-schema Gold layer, and 21 SQL data-quality checks; found that 47.5% of completed orders had totals that did not match their line items, rebuilt revenue from line prices, and cut query time by **25%**.

Medical Cost Modeling & Prediction | *Python, scikit-learn, R* — Oct 2026

- Extended my 2021 R study of 1,338 insurance records: a smoking × obesity interaction explains the three cost bands, raising R^2 from **0.09 to 0.86** and showing the earlier sex effect was confounded by smoking (it flips from +$1,539 to −$494).
- Compared 9 models with 5-fold CV; the interpretable linear model matched ridge, lasso, and random forest and beat boosting (**held-out R^2 0.88**), but subgroup analysis found 29% relative error for non-smokers vs. 5% for obese smokers.

AI, Systems & Theory Coursework | *Python, C++, RISC-V* — 2024 – 2026

- Implemented uninformed and informed search, constraint satisfaction, logical inference, and Bayesian networks in Python, and solved a Connect-4 board puzzle with state-space search.
- Implemented k-means and gradient-descent linear/logistic regression from scratch in Machine Learning (Steam games; 13.6K Melbourne listings); re-tested the housing data with gradient boosting on held-out regions (error 13% → 18–38%).
- Designed a CPU in Logisim, wrote recursive RISC-V assembly, coded DFAs/NFAs, and benchmarked a red-black tree in C++.

PROFESSIONAL EXPERIENCE

Data Analyst Fellow | EDC Lancaster County (Center for Regional Analysis), *Lancaster, PA* — Aug 2026 – Present

- Automated 25+ federal economic data series (BLS, BEA, Census, USPS) into Power BI and Google Apps Script dashboards with daily refresh, saving **15+ hours per month**.
- Embedded a **Gemini LLM assistant** in a Google Apps Script dashboard to summarize trends and flag anomalies for 6+ nontechnical staff.

Data Science Intern | KCC Capital Partners, *Sheridan, WY* — Jan 2026 – May 2026

- Applied statistical and machine-learning models to data on **100+ mining and precious-metals companies** to shortlist 100+ potential buyers for mandates, and automated weekly reports, saving 5+ hours per week.

Data Engineer Intern | KPIM Joint Stock Company, *Hanoi, Vietnam* — Jan 2021 – Apr 2021

- Engineered Python and SQL Server ETL pipelines that automated legacy client data workflows, reducing processing time by **30%**.

SKILLS & CERTIFICATIONS

Programming: Python (Pandas, NumPy, scikit-learn, Matplotlib, PySpark), SQL (T-SQL, Spark SQL), R, C++, Java, C#, RISC-V Assembly, JavaScript
ML & Statistics: Regression, statistical inference, probabilistic reasoning (Bayesian networks), machine learning, model evaluation
Data Systems: Microsoft Fabric (Lakehouse, Dataflow Gen2, Notebooks), SQL Server, Power BI, ETL, data modeling, Git
Certifications: Microsoft Certified: Fabric Data Engineer Associate (DP-700), Oct 2026; HackerRank SQL (Advanced), Sep 2025

SERVICE

Service Year Participant | The Shalom Project, *Lancaster, PA* — Aug 2026 – Jul 2027

- Full-time service year placed at the Center for Regional Analysis, living in an intentional community with a cohort of young adults.